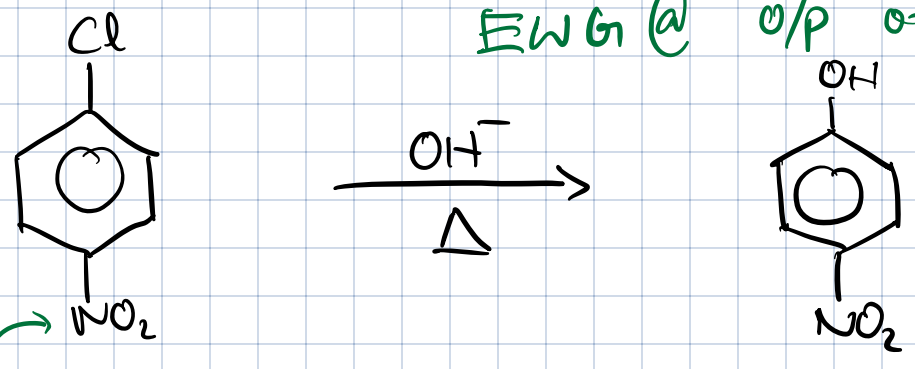
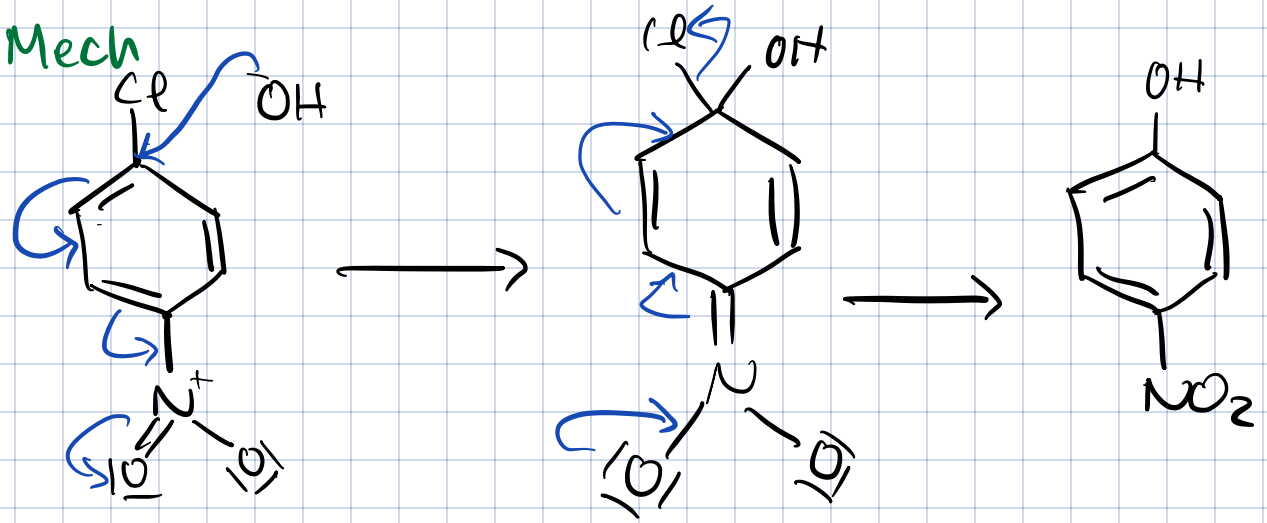
EWG @ o/p of LG

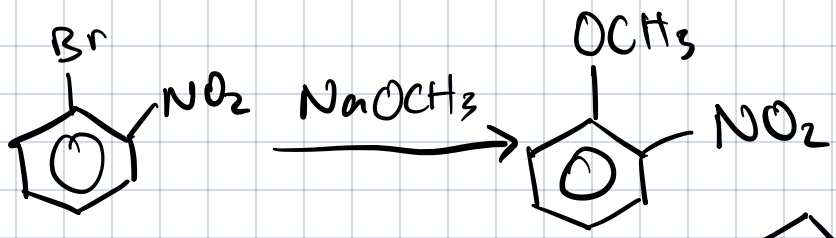


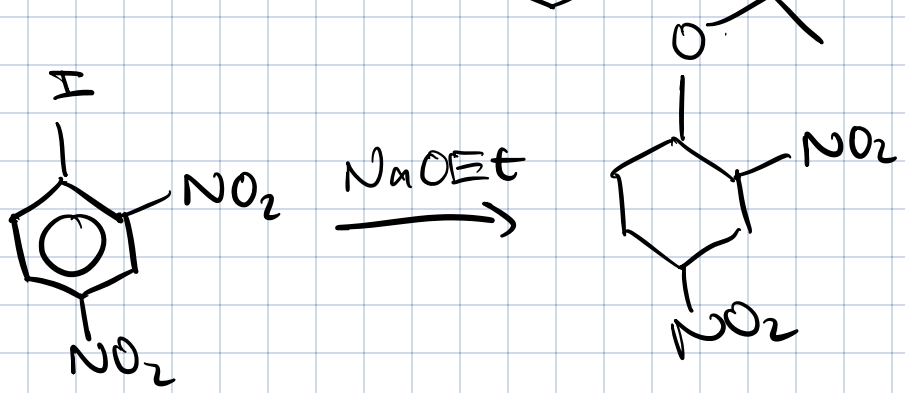
EWG activates SNAR

Mech

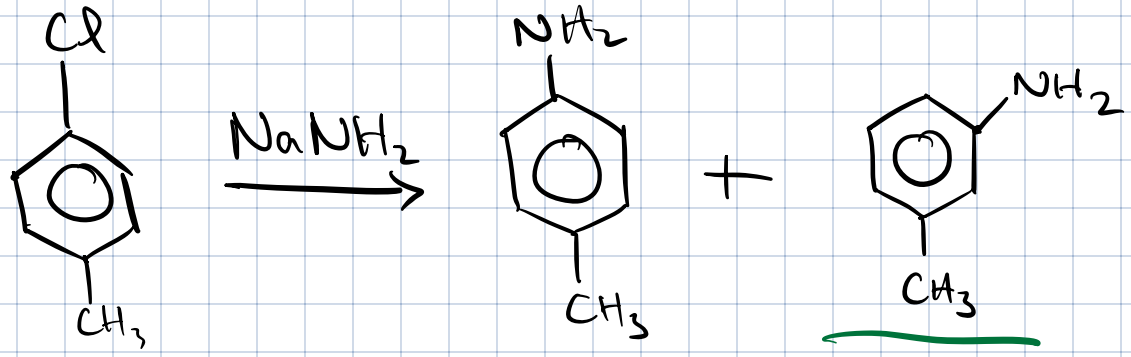


Meisenheimer



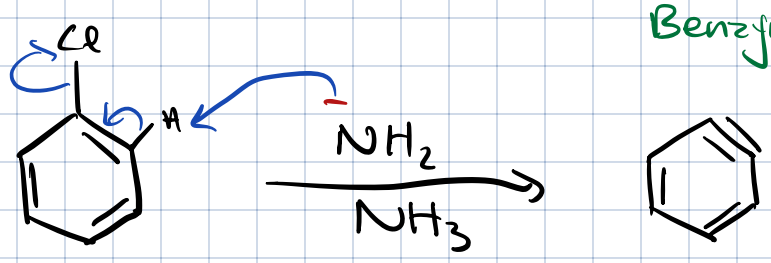


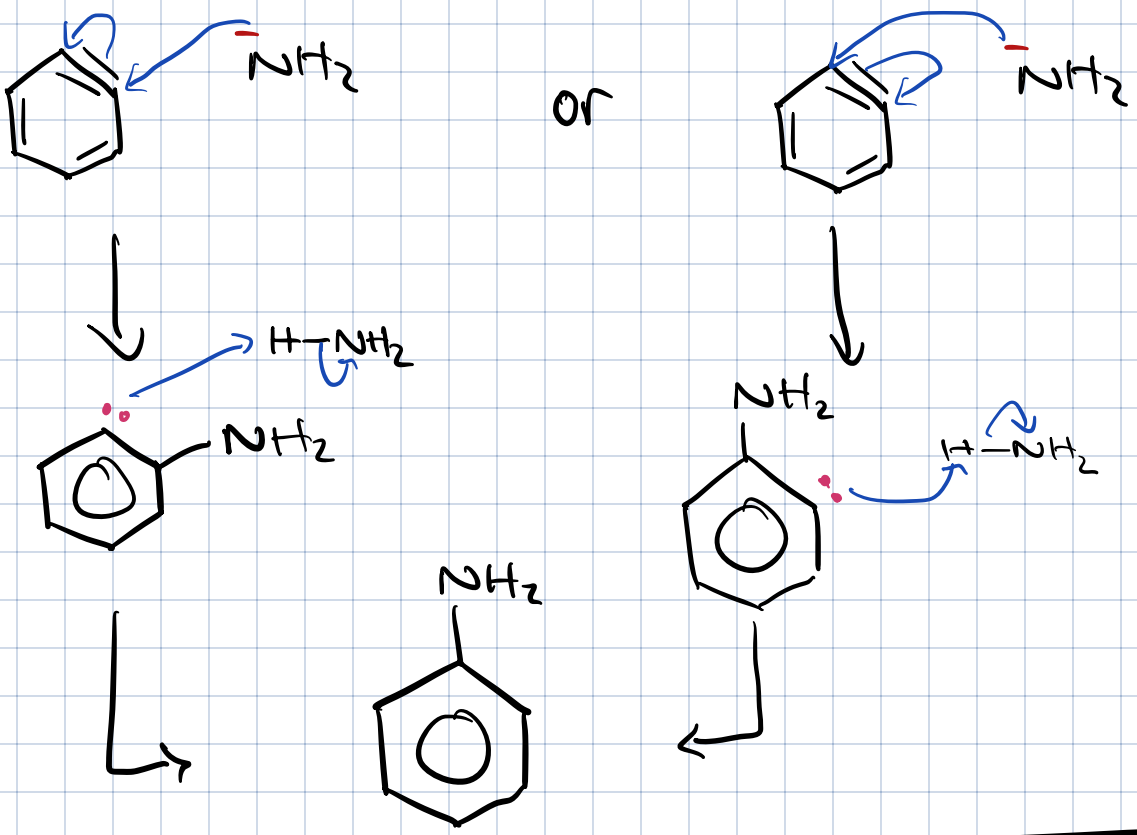
No EWG



Mech w/ no EWG

Benzyne Intermediate





Leaving Group Favorability

F > Cl > Br > I

opposite of SN1/SN2

why?

In SN1 & SN2 leaving group dominates.

F^- is a bad LG.

SnAR is an addition elimination, so LG can stay for the intermediate. The main factor is e⁻philicity of the carbon. Fluorine's −I effect is strongest.

→ See notes above

– Extension: Why is the comparison for F vs. Cl opposite between BF_3 vs. BCl_3 and SNAr

→ Because is SNAr, +M is delocalized but −I is strongest at the substitution carbon, so −I dominates